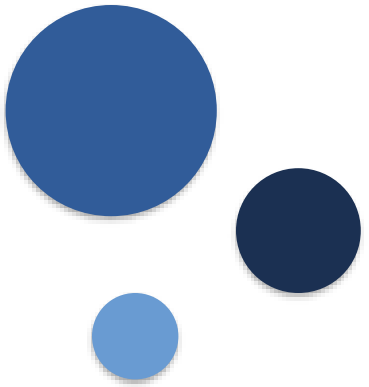
Regular Expressions for Real-World Text Processing

Pattern Matching and Information Extraction



DSA0311 – Natural Language Processing

Technical Presentation
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Introduction to Regular Expressions

A simple way to find patterns in text

A regular expression, or regex, is a sequence of characters that describes a text pattern.

It can search, validate, and extract matching information from large amounts of text.

Regex is supported by languages such as Python, Java, JavaScript, and R.

Example: the pattern \d\d\d matches a three-digit sequence such as 482.

Main idea: describe the pattern once, then let the computer find matching text.

Common Regex Symbols

Symbols help define what kind of text should match

\d	Any digit	7 or 2026
\w	A letter, digit, or underscore	student_1
\s	Whitespace such as a space or tab	Between two words
.	Any character in many regex engines	a, 7, or !
+	One or more repetitions	\d+ matches 482
{n}	Exactly n repetitions	\d{10} matches a 10-digit number

Applications of Regex

Useful in everyday software and data workflows

Data validation: check email, phone number, and ID formats.

Information extraction: collect dates, prices, order numbers, and hashtags.

Log analysis: locate timestamps, IP addresses, error codes, and warnings.

Information retrieval: search files and documents for specific patterns.

Text cleaning: identify repeated spaces or unwanted characters before analysis.

Advantages and Limitations

Use regex where pattern matching is the right fit

Advantages

Fast for pattern-based searches

Reduces manual text checking

Can extract useful data automatically

Limitations

Complex patterns can be hard to read

May miss unusual or unexpected formats

Does not understand meaning or context

Best practice: test patterns with valid, invalid, and unusual examples.

Conclusion

Pattern matching and information extraction

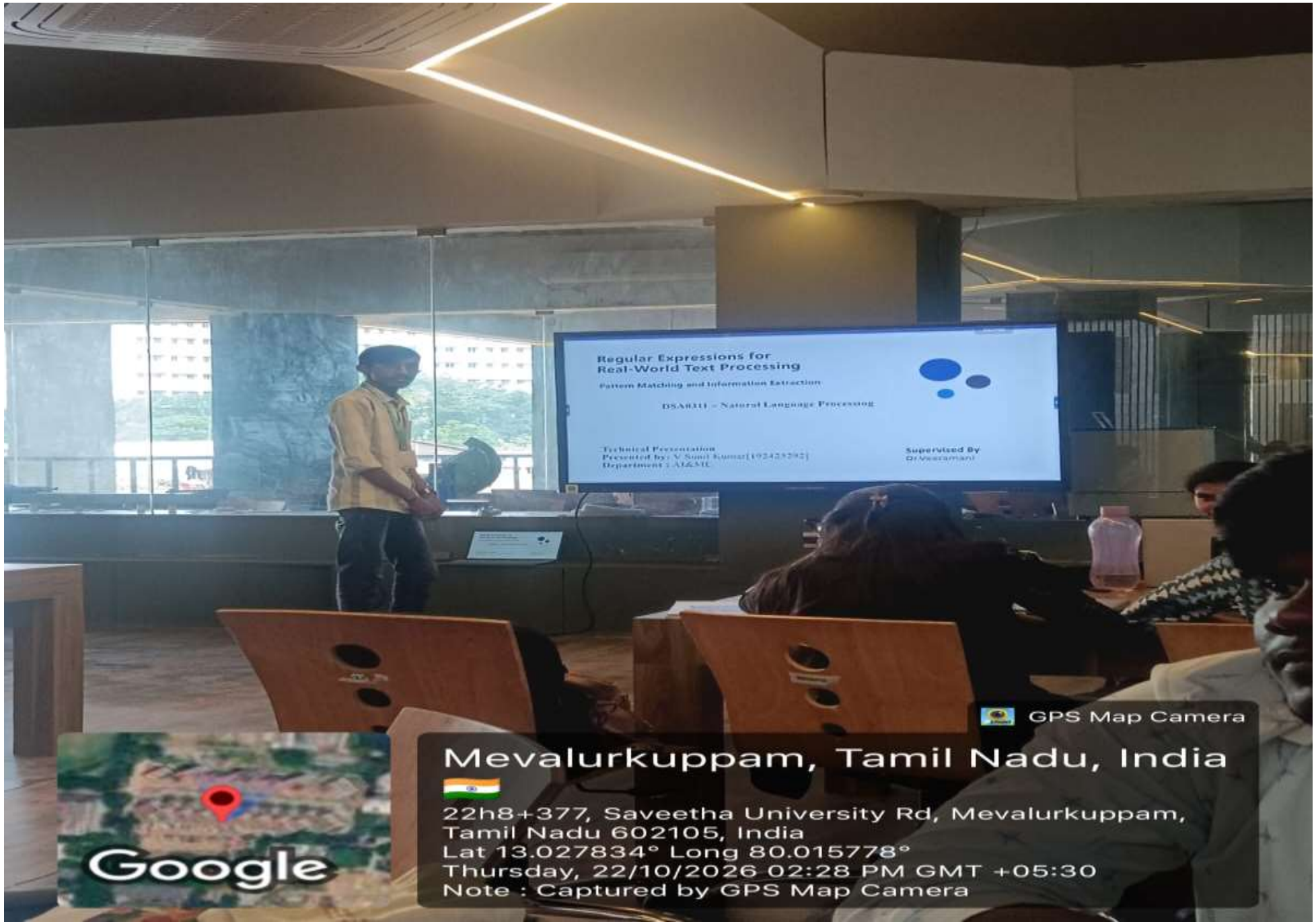
Regular expressions provide a practical way to validate, search, and extract text.

They are useful for emails, dates, phone numbers, web addresses, and system logs.

Clear patterns and careful testing help reduce errors.

For complex structured data or meaning-based tasks, use a suitable parser or other tool.

Thank You


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